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RESEARCH ARTICLE

Assessing the not-invented-here syndrome: Development and validation of implicit and explicit measurements

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Summary

The not-invented-here (NIH) syndrome has been called one of the largest obstacles in innovation management, preventing effective knowledge transfer between organizational units and individuals. NIH is defined as a negatively shaped attitude towards knowledge that has to cross a disciplinary, spatial, or organizational boundary, resulting in either its suboptimal utilization or its rejection. Our goal is to equip scholars with appropriate measurement instruments for the phenomenon. On the basis of 4 studies with 1,238 subjects overall, we developed an implicit measure based on the implicit-association test as well as an explicit (survey) measure of NIH, taking into account theoretical insights on attitude structure. We provide evidence for reliability as well as construct and criterion validity. We want to facilitate further research on NIH and knowledge transfer (a) by providing a better theoretical framework for NIH on the basis of the tripartite componential model of attitudes, (b) by demonstrating the application of association-based implicit measures for management research, and (c) by providing a validated multidimensional survey scale to measure NIH explicitly. We also provide recommendations on how managers can utilize the NIH measurement instruments to investigate NIH and potential countermeasures in detail and they can test the behavioral outcomes postulated by previous research.

KEYWORDS

attitudes, implicit-association test (IAT), scale development, knowledge transfer, not-invented-here

1 | INTRODUCTION

In 1986, Northrop Corporation, a U.S. defense company, terminated its Tigershark program developing the military fighter F-20 without having sold a single plane. The F-20's entire research and development (R&D) investment of over $1.2 billion was privately funded by Northrop and a few subcontractors, a unique situation compared to the usual share of R&D expenditures between a contractor and the U.S. military. As Martin and Schmidt (1987) examine, one of the reasons for the program's failure was the strong negative attitude of the U.S. military, who did not consider the program as one of their own, but instead as the initiative of an outside corporation. For example, the "stars and bars" insignia was removed from an F-20 prototype at the 1984 Paris Air Show because the plane was not a part of the U.S. inventory. Similarly, Northrop was not permitted to land the F-20 at restricted air bases. Northrop management also alleged that the F-20 was not given equal briefings to foreign nations interested in purchasing U.S. fighter aircraft (Martin & Schmidt, 1987).

The rejection of the F-20 as a technological innovation is an excellent example of a phenomenon called the "not-invented-here" (NIH) syndrome, and many managers will recall a similar story from their own experience. Although literature on innovation frequently highlights the importance of incorporating different perspectives, ideas, and technologies into the R&D process (Bogers, Afuah, & Bastian, 2010; Cassiman & Veugelers, 2006; Laursen & Salter, 2006; Wassmer, 2008), innovation as well as knowledge exchange and transfer is often impeded at an individual level (Miron-Spektor, Paletz, & Lin, 2015). This individual barrier, which is referred to as NIH, has been called one of the largest obstacles in innovation management. It is accused of leading to incorrect evaluations and delayed and distorted transfers of ideas and technologies (Agrawal, Cockburn, & Rosell, 2010; Barkema & Schijven, 2008; de Burcharth, Knudsen, & Søndergaard, 2014), slowed implementation and expanded development costs (Lichtenthaler & Ernst, 2006), project failure (Herzog & Leker, 2010; Kathoefer & Leker, 2012), and diminishing firm performance (Katz & Allen, 1982; King, Covin, & Hegarty, 2003). NIH describes "the negatively shaped attitude of an individual toward knowledge that has to cross a contextual (disciplinary), spatial, or organizational (functional) boundary, resulting in either its suboptimal utilization or its rejection" (Antons & Piller, 2015, p. 197).

The term NIH has been widely used in innovation research (e.g., de Burcharth et al., 2014), strategy (e.g., Laursen & Salter, 2006), human resource management (e.g., McKinlay & Starkey, 1992), and marketing (e.g., Franke, Keinz, & Steger, 2009; Hauser, Tellis, & Griffin, 2006). A recent review showed that NIH is referred to in approximately 700 scholarly papers published across various management disciplines (Antons & Piller, 2015). However, this same review also confirmed earlier assessments that literature on NIH has been mainly anecdotal, taking the phenomenon for granted, and discussing its underlying mechanisms only superficially (Agrawal et al., 2010; Kathoefer & Leker, 2012). Hence, the relevance of NIH in both managerial practice and the management literature has not been matched by dedicated research focused on this individual phenomenon of organizational behavior in an innovation context.

We argue that this lack of dedicated research corresponds to a lack of theoretical foundation of NIH. The results are measures and operationalizations of NIH in the previous literature that are not solidly grounded and appear to be used due to the specific purposes or data available. Our argumentation is in line with prior accounts highlighting that the current approaches towards measuring NIH are only in their infancy (Agrawal et al., 2010; Laursen & Salter, 2006).

In their review, Antons and Piller (2015) list all studies addressing NIH as their central construct. They show that prior research has used various proxies to measure NIH. Strikingly, some of these studies do not even measure NIH at all, yet focus on potential outcomes such as performance or communication decline (Allen, Katz, Grady, & Slavin, 1988; Katz & Allen, 1982), patent self-citations in regions (Agrawal et al., 2010), and occurrence of internal resistance against innovation projects (Hussinger & Wastyn, 2011) without establishing the causes behind these effects. Only a few papers that Antons and Piller (2015) reviewed have developed scales more appropriate to capture NIH. For example, Kathoefer and Leker (2012) measure the reluctance to apply external knowledge. This, however, is a measure of behavior induced by the attitude but not a scale measuring the attitude underlying NIH. Similarly, de Burcharth et al. (2014) surveyed R&D managers whether their staff members had negative attitudes towards external knowledge. To the best of our knowledge, only one study tried to develop a scale directly aiming at measuring the attitude underlying NIH (Herzog & Leker, 2010). This study, although citing a definition of NIH as an individual-level construct, conceptualizes the phenomenon as a department-level phenomenon and operationalizes it accordingly. Taken together, a clear and compelling theoretical underpinning is missing that allows developing a measure that follows the definition and conceptualization of NIH as an individual-level attitude (Agrawal et al., 2010; Antons & Piller, 2015; Laursen & Salter, 2006). But without such reliable and valid measures, any future investigations of the antecedents and consequences of NIH and countermeasures against this phenomenon are severely handicapped.

We argue that the development of such a measurement instrument should build on theoretical insights from psychology regarding attitude structure, incorporating the various facets of an attitude (Eagly & Chaiken, 1993; Rosenberg & Hovland, 1960). Thereby, a newly developed measure should assess the NIH attitude holistically incorporating all of these facets. Moreover, psychological attitude scholars distinguish between consciously (explicit) and more automatically processed (implicit) components of an attitude (Fazio & Olson, 2003; Greenwald & Banaji, 1995; Harms & Luthans, 2012; Leavitt, Fong, & Greenwald, 2011). Survey scales are better equipped to measure the conscious components and are, thus, potentially affected by biases due to social desirability. Research on NIH has not taken advantage of more recent findings from psychology on measuring implicit attitude components (Fazio & Olson, 2003; Greenwald, McGhee, & Schwartz, 1998) to validate attitude scales and to measure the phenomenon at hand holistically.

The contribution of this article is fourfold. First, we review literature on NIH in general and the current state of NIH measurement in particular. We introduce findings from psychological attitude research on attitude structure as well as the concept of implicit social cognition and its relevance for attitude measurement. With these understandings, we contribute a theory-driven operationalization of the NIH attitude. Second, with this conceptual understanding of NIH as having an implicit component, we apply a purposefully developed measure based on the so-called implicit-association test (IAT; Greenwald et al., 1998). By doing so, we extend prior effort in showcasing the adaption of implicit measures, and the IAT in particular, to organizational research (Derous, Ryan, & Nguyen, 2012; Harms & Luthans, 2012; Leavitt et al., 2011). Third, in order to reflect the explicit component of NIH, we apply standard scale development procedures (DeVellis, 2011; Hinkin, 1995, 1998) to develop a NIH scale, taking into account theoretical insights from psychology on attitude structure. Fourth, we provide evidence for discriminant and criterion validity of the developed scale and for convergent validity of both the implicit and explicit measures. Together, this allows future research to investigate NIH and potential countermeasures in greater detail, and to test the behavioral outcomes postulated by numerous studies (e.g., Gesing, Antons, Piening, Rese, & Salge, 2015; Laursen & Salter, 2006).

2 | CONCEPTUAL BACKGROUND

2.1 | The not-invented-here syndrome and its underlying attitude

As our introductory example highlights, NIH is an attitude-induced decision-making bias that occurs during the evaluation of knowledge from origins being external due to contextual (disciplinary), spatial, or organizational (functional) boundaries (Antons & Piller, 2015; Gesing et al., 2015; Kathoefer & Leker, 2012). The rejection of external ideas, technologies, and knowledge is no problem per se; when the external input is of no value for the absorbing organization and its members, staff act wisely by selecting input critically. From an organizational point of view, rejection or underutilization of external input is damaging in cases where input would provide value in comparison to internal solutions or when knowledge is rejected due to attitudinal biases rather than actual facts (Kathoefer & Leker, 2012).

Not-invented-here is defined as an attitude-based decision bias. According to social psychology, attitudes are a major factor in human interaction and decision making (Eagly & Chaiken, 1993). Attitudes guide an individual's thinking and information processing to the extent that attitude- and belief-consistent information is sought (Munro & Ditto, 1997; Petty & Wegener, 1998). Hereby, they influence decisions and thus behavior (Ajzen, 2001). In doing so, they fulfill five functions for an individual by fostering and maintaining self-esteem and self-concept, expressing and affirming an individual's values, facilitating and maintaining social relations by advancing social identities, structuring and organizing the perception of the environment, and helping to evaluate information during decision making (Ajzen, 2001; Eagly & Chaiken, 1993).

Consider the example of a software developer, who has to evaluate new product ideas provided by a user community. Some of the ideas may be unfeasible or unaffordable. Here, rejection is purely rational without being biased due to NIH. For other ideas, however, an attitude might trigger rejection. Because users in online forums often describe their ideas in rather colloquial language, the developer might view them as unprofessional and thus not worth considering more thoroughly. In addition, the developer might conceal other good ideas out of fear that a superior criticizes her or him for not being creative herself or being dependent on external input. Together, an attitude maintaining the self-concept ("I am the expert") and shielding the developer from expected negative consequences might bias the evaluation of such ideas, eventually leading to rejection due to NIH.

2.2 | The tripartite componential model of attitudes

Early attitude research has established a tripartite componential model of attitudes (Rosenberg & Hovland, 1960). According to this model, attitudes encompass affective, cognitive, and behavioral components (Eagly & Chaiken, 1993). The affective component reflects feelings, moods, and emotions towards the attitude objective. Opinions, thoughts, and ideas are condensed into the cognitive component. Finally, the behavioral—or sometimes called conative—component embraces views on prior behavior and actions as well as former behavioral intentions towards the attitude object. Although the model was not consistently supported in early empirical studies, more sophisticated research starting in the 1980s has confirmed it strongly (e.g., Breckler, 1984). Nowadays, it is commonly accepted (Ajzen, 2001; Bohner & Wänke, 2002; Eagly & Chaiken, 1993) and serves as a theoretical reference point for attitude scale development.

Although not referring to the tripartite componential model, prior NIH studies implicitly adopted this structure. For example, Katz and Allen (1982) state that R&D staff members "become increasingly committed to their current problem-solving strategies, their customary ways of doing things, and their traditional modes of conduct" (p. 18). In a similar vein, Antons and Piller (2015) describe that *behavioral* customs form attitudes:

> *Expert tracks, the notion of "innovation champions," or an award culture recognizing expert achievements (such as Procter & Gamble's famed Victor Mills Society) foster the construction of pockets of expertise held by one individual. Such individuals may demonstrate NIH when confronted with external knowledge in their particular domains to defend their expert status (p. 200).*

With regard to the *affective* attitude component, Lüttgens, Pollok, Antons, and Piller (2014) describe a fear of being viewed as incompetent: "Why should we find an external solution when we can do it ourselves? If we do this, we're really only showing how bad we are" (p. 552). Similarly, Kathoefer and Leker (2012) propose that "the integration of external knowledge may injure a research group's pride. People may feel affronted when external ideas are regarded as superior to their own" (p. 660).

Finally, the cognitive component mirrors opinions and thoughts towards the attitude object. Burrows and Greene (2000) describe the reality-distortion situation at Apple in the 1990s that led them to reject external ideas. Citing from an interview with Steve Jobs, they write: "Making the iMac work without a noisy fan is not just fit-and-finish. It is hard-core engineering. […] There is not another PC company in the world that could have designed the Cube. They don't have the engineering, nor do they have the design talent" (p. 112). This statement reflects the opinion that Apple had superior design capabilities at that time. Due to this widely held belief, external ideas were not considered sufficiently. Likewise, Zak (2013) describes how Twitter initially rejected to use the hashtag symbol to channel tweets about a specific event. Although being used by many users at this point, the Twitter development team considered the hashtag to be "too nerdy" and not appealing to a wider audience. This policy was only changed after users continued using the hashtag symbol. Hence, viewing the tag as nerdy and unprofessional led to rejection despite the benefit it provided for users in channeling communication.

As an illustration of the tripartite componential structure in the context of NIH, consider the example of technologies such as geological disposal of CO_2 or artificial intelligence. These developments are accompanied by wide public discussions of dreaded negative outcomes. The individual developer who is expected to adopt such technology will certainly also have an opinion on, for, and against arguments, will perhaps share the fear of the possibly negative outcomes, and may have some personal experience of the technologies, perhaps gained through the academic study of laboratory experiments. Together, these thoughts, fears, and behavioral experiences make up the three components of the model and thus may form an individual's attitude towards the respective technology. As this example implies, the components may co-occur, whereas other attitudes may also be formed on the basis of only a single component (Ajzen, 2001; Eagly & Chaiken, 1993).

As such, NIH research is well advised to reflect this tripartite componential structure to measure the phenomenon holistically. If an individual's attitude is based on a single component and this component is not reflected in the actual measure, the researcher will not be able to assess the phenomenon and to answer the research question underlying the measurement. Moreover, having measured all three components separately, scholars are able to identify those factors that determine the three components and distinguish which component drives rejection of external knowledge under which circumstances. As the above example implies, the affective component might be a much stronger driver of rejection of external knowledge in environments where employees feel the pressure to compete for promotion on the basis of their own expertise.

2.3 | The differentiation of implicit and explicit attitudes

During the last two decades, attitude theorists have begun to distinguish between two different conceptualizations of attitudes (and cognitive processes in general). Coined by Greenwald and Banaji (1995), the term "implicit social cognition" refers to cognitive processes occurring without conscious awareness or control. This includes social psychological constructs such as attitudes, stereotypes, and self-concepts. Regarding the evaluation of external knowledge, Menon and Blount (2003) describe that personal relationships with the messenger of the knowledge might become an implicit characteristic or attribute of the knowledge. Using the relationship as a cue, the attitude towards the knowledge is formed implicitly and in an automated manner. Implicit attitudes are thus seen as an automatic evaluative reaction that is present in an individual's mind when noticing a specific attitude object (Fazio & Olson, 2003). In contrast, people are aware of explicit cognitions and might, thus, control them. Explicit attitudes are thus a more deliberative assessment of the attitude object that requires reflection.[1]

To distinguish between implicit and explicit components of an NIH attitude, consider again our previous example of a software developer, who has to evaluate new product ideas provided by a user community. Similar to Twitter's initial rejection of the hashtag as a tool to channel user communication because it "looks too nerdy" (Zak, 2013), the developer might reject those ideas being explicated in rather colloquial language in an intuitive manner viewing them as unprofessional and thus not worth considering more thoroughly. This might reflect a more automatic evaluation and, thus, rejection triggered by a more implicit attitude. The developer's rejection of good ideas out of fear that a superior criticizes her or his for not being creative might be seen as more deliberate.

2.4 | An overview of implicit attitude measurement

Implicit measures have been developed to describe implicit social cognition and, hence, to tap into the reasons for individual perceptions, judgments, or actions, even in instances where respondents lack the ability to think self-reflexively or the motivation to report accurately (Fazio & Olson, 2003; Nosek, Hawkins, & Frazier, 2011). Previous methodological literature has emphasized potential avenues for the application of implicit measures in management research (Haines & Sumner, 2006; Uhlmann et al., 2012).

[1]Reviews of psychological research on attitudes highlight that attitude research is still in its infancy in understanding whether the explicit–implicit distinction taps into the same attitude, different aspects of dimensions of attitudes, or different processes activating attitudes (Bohner & Dickel, 2011; Crano & Prislin, 2006; Fazio & Olson, 2003). As one anonymous reviewer highlighted, this implies that the linkage between the tripartite componential model of attitudes (Rosenberg & Hovland, 1960) and the explicit–implicit distinction is in strong need for future theory building.

The IAT, introduced by Greenwald et al. (1998), is the most prominent implicit measurement instrument (Lane, Banaji, Nosek, & Greenwald, 2007; Nosek et al., 2011; Uhlmann et al., 2012). In the Uhlmann et al. (2012) terms, the IAT is an association-based implicit measure. This group of implicit measures aims to assess the links between target concepts and their attributes, often by triggering automatic responses (Fazio & Olson, 2003; Nosek et al., 2011; Uhlmann et al., 2012). These instruments share the underlying assumption that categories are represented in an individual's knowledge structure. Activation of one category activates other related categories (Greenwald et al., 2002). For example, if an engineer has a negative attitude towards marketers, activating the concept "marketers" would also put into play concepts such as negative affect and negative cognitions. Measuring the strength of these associations typically relies on reaction times to stimuli that represent the different target categories of the specific test, which are displayed in rapid succession. The measures usually include a within-subject experimental design that compares performance between different test conditions (Nosek et al., 2011; Uhlmann et al., 2012) to tap into reaction time differences and thus differences in association strength.

The IAT provides a relative measure, comparing two target categories and their associations with two attribute categories. A scholar interested in researchers' perceptions of knowledge from different disciplines, such as management and mechanical engineering, might be interested in stereotypes, such as *management + high-performing* or *management + low-performing*, or attitudes, such as *mechanical engineering + positive* or *mechanical engineering + negative*. Applying a within-subject test, the IAT provides a relative measure, comparing the combination of *mechanical engineering + positive* and *management + negative* with the combination of *mechanical engineering + negative* and *management + positive*. When subjects react faster to stimuli representing the categories of the first combination than they do for the second one, the IAT results are interpreted as demonstrating greater association strengths between the target categories and the attributes in the first combination. More specifically, this means that the individual favors mechanical engineering over management science.

Unlike some other implicit measures, the IAT meets psychometric criteria such as reliability (Fazio & Olson, 2003; Hofmann, Gawronski, Geschwendner, Le, & Schmitt, 2005; Lane et al., 2007) and predictive validity (Greenwald, Poehlman, Uhlmann, & Banaji, 2009; Hofmann et al., 2005). However, it has been criticized for being relative, meaning the IAT is not able to distinguish attitudes at the level of single categories and can only evaluate categories in relationship to each other (Fazio & Olson, 2003; Nosek & Banaji, 2001; Uhlmann et al., 2012). Thus, it is not possible to investigate whether the attitudes are reciprocal or whether they are independent from each other. Still, although many other implicit measures have been developed in social cognition research, the IAT is, nevertheless, preferred due to its established quality.

2.5 | Measuring the NIH attitude

Measures of NIH used in articles already published can be put into two categories. First, most articles observe behavioral effects, such as refusing behavior or declining performance, and state that they result from NIH without providing a measure of the underlying attitude leading to the behavior (Agrawal et al., 2010; Katz & Allen, 1982). Second, a few recent papers provide surveys aiming at measuring NIH. They can be subdivided into single items investigating the existence of NIH in an organization in order to measure, for example, organizational openness to external input (e.g., Fey & Birkinshaw, 2005), and scales aiming at measuring NIH as a construct of its own (de Burcharth et al., 2014; Herzog & Leker, 2010; Kathoefer & Leker, 2012). However, instead of measuring the NIH attitude, these scales measure outcomes such as the reluctance to apply external knowledge (Kathoefer & Leker, 2012), are operationalized on a department level rather than an individual level (Herzog & Leker, 2010), or use a third-person approach surveying managers on the attitudes of their staff members (de Burcharth et al., 2014).

From a conceptual point of view, all of these scales disregard the established threefold structure of attitude components (affective, behavioral, and cognitive; Rosenberg & Hovland, 1960). Rather, the existing NIH scales do not measure the attitude at all but its behavioral outcomes or focus on either the behavioral or cognitive attitude components and contain questions regarding the utilization of external knowledge, collaboration with external partners, or trust in internal versus external knowledge.

From a methodological point of view, the existing survey measures have in common that they use Likert scale items organized in a standardized questionnaire. These measures ask direct questions, a method that renders them transparent, and means that the interviewee has introspection in the aim of the measurement and is able to adjust his or her answer accordingly. These types of measures are often considered as explicit and being mainly able to tap into the explicit component of an attitude (Fazio & Olson, 2003). However, questions related to attitudes are often sensitive to response bias: In times of open innovation, utilizing external knowledge and technologies is often promoted as being positive, and as this might be contrary to an individual's attitude, questions might be answered in a biased manner in order to meet external expectations. Hence, more sensitive issues are often concealed in attitude surveys due to social desirability and self-expression (Fazio & Olson, 2003; Uhlmann et al., 2012). Moreover, regarding common psychometric criteria such as reliability and validity, prior literature suggesting attitude scale for NIH provide little to no information or exhibit low to modest quality (e.g., de Burcharth et al., 2014; Kathoefer & Leker, 2012).

Taken together, NIH measurement can be viewed as still in its infancy. Available measures do not include theoretical insights regarding attitude structure or do not measure the attitude at all, exhibit low to modest psychometric quality, are potentially affected by biases due to social desirability, and are not able to account for the implicit component of NIH attitudes. Hence, scholars interested in understanding NIH, its antecedents, consequences, and countermeasures need new and appropriate explicit as well as implicit measures. In the following, we will present a set of studies dedicated to developing these measures and incorporating both implicit (Study 1) and explicit measures (Study 2). We will investigate the interrelation of the two NIH measures, analyzing the convergent validity of both measures (Study 3). Finally, we provide evidence for discriminant and criterion validity of the explicit NIH measure (Study 4). As highlighted by earlier research, NIH may occur due to contextual, organizational, and spatial boundaries that the external input has to overcome (Antons & Piller, 2015). To focus our research, we investigate just one specific boundary—the bridging of contextual or disciplinary boundaries—leaving it to future studies to conceptualize the other boundaries.[2] We believe that our focus on disciplinary boundaries is timely and supported by today's relevance of interdisciplinarity for the advancement of science and academia (Oppenheimer, 2015; Rinia, Van Leeuwen, Bruins, Van Vuren, & Van Raan, 2001; Starkey & Madan, 2001).

3 | STUDY 1—IMPLICIT ATTITUDE MEASUREMENT

The first study was conducted to develop an NIH-related IAT and demonstrate its ability to capture the NIH attitude with regard to different knowledge disciplines. For the focus of the research, we chose the two knowledge domains of mechanical and electrical engineering, which are becoming more and more interconnected, especially in the growing field of mechatronics. In our study, mechanical and electrical engineers responded to stimuli representing the two disciplines (the "target concepts") as well as to words representing positive and negative attribute categories. Following the standard assumption of an IAT, we expected faster responses for compatible combinations (*own knowledge domain + positive* and *other knowledge domain + negative*) than for noncompatible ones (*other knowledge domain + positive* and *own knowledge domain + negative*).

3.1 | Method

3.1.1 | Participants

The sample for this first study consisted of 42 graduate students (average actual duration of study of 3.67 years; average age of 23.9 years, ranging from 19 to 32 years) from a large German university of technology, and 46 professionals with graduate degrees (average work experience of 5.77 years; average age of 31.3 years, ranging from 25 to 43 years) working in different technical departments of an engineering company (contract research). Of these 88 individuals, 38 had a background in electrical engineering and 50 in mechanical engineering. The electrical engineering group consisted of 15 students (one female) and 23 male professional developers. The mechanical engineering group was made up of 27 students (two females) and 23 professionals (three females).

[2]Stimulated by the advice of one anonymous reviewer, we surveyed 352 European innovation scholars and practitioners to assess their agreement to the definition and conceptualization of NIH drawn from Antons and Piller (2015). We received 105 completed responses to four questions on the agreement to the entire definition and its three components that (a) NIH is based on an individual attitude; (b) the object of the attitude is knowledge crossing a contextual, spatial, or organizational boundary; and (c) the attitude results in the suboptimal utilization of the external knowledge or its rejection. The respondents expressed strong support for the definition. On a 7-point Likert scale, the average agreement was 6.1 (standard deviation 0.97). Agreement to the three components ranged from 6.0 to 6.2.

3.1.2 | Materials

To represent the two knowledge domains of mechanical and electrical engineering, appropriate stimuli had to be found. According to Lane et al. (2007), stimulus identification and selection is crucial for adapting the IAT to the construct and categories at hand. Thus, prior to the actual IAT study, we developed stimulus lists for the target categories with the help of 55 individuals who did not take part in the later study. In a first step, 15 individuals were asked to identify words that are representative of the domains of mechanical and electrical engineering, resulting in a consolidated list of over 75 words. In a second step, 40 new individuals rated all the words on a 5-point Likert scale with regard to their representativeness for both mechanical and electrical engineering, respectively. Out of the larger list, the 40 stimuli (e.g., gearbox and diode) with the highest ratings were selected for use in the final IAT. Following previous IAT studies (e.g., Greenwald et al., 1998; Hekman et al., 2010; Nosek, 2005), words related to “positive” and “negative” reflected the attribute categories. Hence, 40 attribute stimuli (e.g., cheer and disaster) were randomly selected out of the stimulus set originally used by Greenwald et al. (1998) and translated into their German equivalents.

3.1.3 | Procedure

We followed the original IAT procedure described in Greenwald et al. (1998), as illustrated in Table 1. The task consisted of three practice blocks and two test blocks. In the first two blocks, participants practiced the task and responded only to the target categories (Block 1) and attribute categories (Block 2). All other blocks were combined blocks where target categories and attribute categories appeared in combination, as described above. Reminder labels displaying the category names remained on screen during the block. Categories (Blocks 1 and 2) or category combinations were assigned to the left or right side of the screen, and participants had to respond with the right hand for the right category, and vice versa, using the “e” and “i” key on a qwertz keyboard.

The stimuli were presented in black letters on a light gray background and centered on the computer screen. All stimuli remained on the screen until the subject responded. The interval between a response and the appearance of the next stimuli for all blocks was 400 ms. During the practice blocks (Blocks 1, 2, and 4), 40 trials were carried out. There were an equal number of category stimuli and attribute stimuli within the “category only” block (Block 1) and in the “attribute only” block (Block 2) respectively, whereas in the mixed practice block (Block 4), 10 randomly chosen stimuli represented each category and attribute group. The experimental blocks (Blocks 3 and 5) consisted of 80 trials. For each category and attribute group, all 20 stimuli were presented. The order of all 80 stimuli was random. Each stimulus was presented once during each block. To avoid confounds, the category and attribute label combinations and placement on the screen (left or right) were counterbalanced across participants, resulting in four different variants of the test. The experiment blocks were processed, and the responses recorded using E-prime version 2.0 professional.

Methodological research on the IAT has produced two different algorithms for analyzing the reaction times (RTs) resulting from the test blocks: the conventional algorithm introduced by Greenwald et al. (1998) and the improved algorithm provided by Greenwald, Nosek, and Banaji (2003) resulting in the so-called D measure. Both measures utilize the effect that participants respond faster in the attitude-compatible condition than in the noncompatible condition of the test. The algorithms used to calculate both effect measures are described in Appendix A. Following the procedure described in Greenwald et al. (2003), we calculated a satisfactory split-half reliability (Spearman–Brown coefficient) of 0.77 for the IAT.

3.2 | Results

The difference in response times demonstrated by the D measure and the conventional effect measure is called the IAT effect (Greenwald et al., 1998), which depicts the focused attitude, its dimension (positive or negative), and its magnitude. In our specific case, the IAT effect reflects the severity of NIH towards the knowledge of the other engineering domain. The analysis of the individual D measures ($\mu = 0.348$, $\sigma = 0.360$) revealed that participants responded faster when the categories were compatible (*own knowledge domain + positive* and *other knowledge domain + negative*) than in the incompatible condition, $t(87) = 9.058$, $p < .001$, $r = .697$. The conventional algorithm ($\mu = 0.107$, $\sigma = 0.117$) demonstrated a similarly strong effect, $t(87) = 8.521$, $p < .001$, $r = .674$. Both measures correlated very strongly, $r = .934$, $p < .0001$. Thus, participants with a background in mechanical engineering responded faster when the domain of mechanical engineering was combined with the positive attribute category and when electrical engineering was combined with the negative, compared with the other combinations. Accordingly, having a background in electrical engineering led to a faster response when electrical engineering was pooled together with the positive attribute category.

TABLE 1 IAT design utilized for Study 1

		Block				
		1	2	3	4	5
		Task description				
		Practice of target concept discrimination	Practice of attribute concept discrimination	Combined test block	Practice of reversed combination	Combined test block of reversed combination
Mapping (example)	Left	Mechanical engineering	Positive	Mechanical engineering + positive	Mechanical engineering + negative	Mechanical engineering + negative
	Right	Electrical engineering	Negative	Electrical engineering + negative	Electrical engineering + positive	Electrical engineering + positive

Note. IAT, implicit-association test.

The order and placing of the categories on the screen and the resulting four variants of the IAT did not differ significantly with respect to the D measures, $F(3, 84) = 0.322$, ns. Likewise, the analysis showed no statistical difference between the effects for the students ($\mu = 0.403$, $\sigma = 0.325$) and for the professionals ($\mu = 0.297$, $\sigma = 0.386$), $t(86) = 1.39$, ns.

3.3 | Discussion

In Study 1, we developed an implicit measure for NIH on the basis of the IAT. We operationalized the phenomenon in an interdisciplinary context, measuring the relative attitude towards both mechanical and electrical engineering. From a conceptual point of view, this operationalization of NIH may be viewed as a conservative test as these knowledge domains are rather interrelated and continue to consolidate new fields such as mechatronics. However, we were able to show that individuals favor their own knowledge domain over a different but related domain. Using *r*-family effect sizes, our analysis exhibited rather strong attitude differences with respect to the two knowledge domains. We would expect attitude effects to be even stronger when the attitude categories are less connected than mechanical and electrical engineering.

Importantly, we found no significant differences in the effect sizes between the two experimental groups represented by students and professional developers. In contrast, the existing literature suggests that NIH will increase with tenure and experience (Katz & Allen, 1982). The evidence provided by our results is not in line with this suggestion, which begs the question of how NIH is affected by experience. Beyond offering, to our knowledge, the first dedicated implicit measurement instrument for NIH, Study 1 also provides evidence of the usefulness of IATs for organizational research in general (Harms & Luthans, 2012). Our method description may equip scholars with a valuable starting point when designing their own IAT by describing the development of stimulus lists, the tasks' procedure, and the analytical strategy in some detail. Echoing prior literature (Lane et al., 2007), we recommend pilot testing stimuli choice. Stimuli that may be categorized quickly and easily and are not confounded with other categories will lead to small error variance. Therefore, appropriate stimulus choice maximizes the chance of generating proper results regarding the implicit social cognition that the researcher is interested in.

Applying both available algorithms to calculate IAT effects revealed only minor differences with regard to statistical effects. Yet these minor differences are in line with an extensive study of about 39,000 individuals that tested different ways of computing IAT effect sizes and recommends the D measure due to the superior construction of its algorithm and resulting larger statistical effect sizes (Greenwald et al., 2003).

4 | STUDY 2—EXPLICIT ATTITUDE MEASUREMENT

As stated above, prior research on NIH has just started to operationalize the construct (de Burcharth et al., 2014; Herzog & Leker, 2010; Kathoefer & Leker, 2012). Following the dominant design of empirical organizational inquiry, these studies have applied survey measures, which are labeled as explicit measures in psychological research (Derous et al., 2012; Uhlmann et al., 2012). However, as argued before, these previous NIH survey measures have, in their item design, not included the consensus in psychological research that attitudes are composed of affective, cognitive, and behavioral components (Ajzen, 2001; Bohner & Wänke, 2002; Eagly & Chaiken, 1993; Rosenberg & Hovland, 1960). Hence, we developed a new scale to measure NIH in interdisciplinary settings that takes into account the theoretical structure of attitudes.

4.1 | Method

4.1.1 | Participants

The second study included data from 656 individuals (144 females; average age 26.0 years, ranging from 18 to 56) who took part in the three iterations to develop the explicit scale. Of these, 151 had a background in electrical engineering, 313 in mechanical engineering, and 192 in management.

4.1.2 | Scale development procedure

We followed suggestions from the literature on scale development (DeVellis, 2011; Hinkin, 1995, 1998) to (a) generate items, (b) administer our survey, (c) reduce items by means of item analysis, and (d) provide evidence for the factorial structure using confirmatory factor analysis. With the substantial body of research on attitudes, attitude structure, and explicit attitude measurement (for reviews, see, e.g., Ajzen, 2001; Bohner & Dickel, 2011; Petty, Wegener, & Fabrigar, 1997) as well as the conceptual and phenomenological literature on NIH (Antons & Piller, 2015), we decided to follow a deductive scale development approach (Hinkin, 1998). We generated three subscales to reflect the three attitude components of the tripartite componential model (Rosenberg & Hovland, 1960), as described above. As a first step, we conducted an expert workshop and developed items jointly with two experienced scholars from social psychology, resulting in 29 items. After that, we discussed the formulation (e.g., clarity and simplicity of language, short sentences, and perspective) with a broader group of psychologists as well as management scholars. Finally, we pretested the items with 40 participants to assess content validity. All participants were asked to think aloud and describe their ideas and thoughts regarding the items.

We then set up an online survey, applying a slider in order to measure on a truly metric scale ranging from 0 (*strongly disagree*) to 100 (*strongly agree*). In a first item development iteration, 316 participants answered this survey. Following recommendations from literature on attitudes (Eagly & Chaiken, 1993), we inspected the items on the basis of four common criteria from classical test theory, namely, item difficulty, item homogeneity, item-to-total correlation, and Cronbach's alpha (Cronbach, 1951; Field, 2013; Gilbert & Churchill, 1979; Lienert & Raatz, 1981). This analysis showed that some items needed refinement whereas others were rejected due to bad coefficients or difficulties in understanding reported by the participants. We revised the scales and added some new items to replace the items being filtered out. This second version of the questionnaire contained 23 items and was answered by 124 participants. We repeated the item analysis and again refined the scale.[3]

In a third iteration, 216 participants answered the refined scale drawn from the second iteration. These data were used to perform a confirmatory factor analysis in order to provide evidence for the three-factorial structure that we draw from attitude theory.[4] Table 2 depicts the final items measuring NIH using three subscales pointing to the three single attitude components.[5] All factor item loadings ranged from 0.554 to 0.786 and were significant (p < .001; see Table 2). As indicated in Table 3, the cognitive component has insufficient discriminatory validity with regard to the conservative Fornell–Larcker ratio (Fornell & Larcker, 1981). Hence, we tested a number of alternative models to compare fit indices (Byrne, 2010). Table 4 illustrates fit statistics for the different models. The results support the proposed three-factorial model. This model possesses a good fit compared to the other models with regards to the cutoff criteria for the fit indices (Byrne, 2010; Hu & Bentler, 1999; Schermelleh-Engel, Moosbrugger, & Müller, 2003; Sharma, Mukherjee, Kumar, & Dillon, 2005).

4.2 | Results

To analyze the subscales, we computed the arithmetic mean of all corresponding items. The mean of these three values serves as the NIH score. We analyzed the three subscales and the overall NIH score for the final version of the scale and the 216 participants of the study. Table 5 lists means, standard deviations, and correlations for all measures. The participants reported rather moderate NIH scores. Only with regard to the behavioral subscale was the mean score notably higher. As expected, the three subscales are strongly correlated with each other and the overall NIH score.

4.3 | Discussion

Study 2 developed an explicit measure of NIH rooted in the componential attitude model. We followed standard procedures in attitude scale development. This advances the field as prior explicit measures for NIH were either abstract using measurements such as declining communication rates (Katz & Allen, 1982) or patent self-citations (Agrawal et al., 2010) or, when applying Likert-type scales, neglected the theoretical structure of attitudes and provided little information on scale development, assessment, and quality (de Burcharth et al., 2014; Hussinger & Wastyn, 2011; Kathoefer & Leker, 2012). We hope that the scales developed, tested, and validated in Study 2 for the disciplinary boundary will offer standard scales for future research to measure individual attitudes towards interdisciplinary knowledge exchange. Moreover, the items may serve as a starting point to develop scales for spatial and organizational boundaries, where NIH is also likely to occur (Antons & Piller, 2015).

5 | STUDY 3—CONVERGENT VALIDITY OF NIH MEASURES

The third study was designed to demonstrate convergence of the NIH focused IAT (Study 1) and the survey measure (Study 2). Our motivation was to investigate the convergent validity of these two measurement instruments to first support future NIH research, and also to contribute to the general methodological debate on implicit and explicit measures in organizational research.

Given the growing emphasis on cross-functional collaboration between managers, marketers, and engineers (see Troy, Hirunyawipada, & Paswan, 2008, for a meta-analysis), we decided to broaden the disciplinary scope of Studies 1 and 2 and tap into attitudes regarding the knowledge domains of "management" and "mechanical engineering." Hence, for Study 3, we developed a corresponding IAT, taking these knowledge fields into account.

5.1 | Method

5.1.1 | Participants

The third study includes data from 124 students (46 females), who answered the survey and the corresponding IAT: 63 had a background in mechanical engineering (average age of 24.0 years, ranging from 18 to 33 years) and 61 were majoring in management (average age of 23.9 years, ranging from 18 to 43 years).

5.1.2 | General procedure

Large-scale studies on order effects of explicit and implicit measures (Nosek, 2005; Nosek, Greenwald, & Banaji, 2005) and a meta-analysis (Hofmann et al., 2005) provided evidence that the presentation order of both measures does not influence their outcomes or inflate correlations. However, to avoid any potential biases due to order effects, we applied a delaying measurement (Greenwald et al., 2009) in this study. This entailed the participants answering the explicit attitude measure online at least 7 days before they came to the laboratory to conduct the IAT.

5.1.3 | IAT materials and procedure

In this study, we wanted to represent the knowledge domains of mechanical engineering and management. Although we had already developed a list of stimuli for mechanical engineering in Study 1, we

[3]Appendix B provides an overview of the two iterations of the item analyses. Please note that all coefficients displayed in Appendix B are based on the preliminary version of the scales. In order to improve the quality of the scale, we conducted a third iteration of the scale development process.

[4]Following editorial guidance, we also conducted an exploratory factor analysis yielding three factors. Results are available from the authors upon request.

[5]One anonymous reviewer emphasized that our items are all phrased positively. To gain our NIH score, we inverted all responses. The reviewer pointed out some potential methodological weaknesses that these facts together might induce noise into the measurement. To check whether this positive phrasing disturbs our results, we conducted a test–retest reliability study. We surveyed 79 scientific staff members of a research department of a large German university yielding 60 complete responses. All participants answered our scale as reported in Table 2 and a negatively phrased version of all items in a second part of our study. Canonical correlation analysis revealed a coefficient of 0.768 and the test of Wilk's λ with a coefficient of 0.410 indicates high significance of the correlation, $F(1, 58) = 83.671$, p < .001. The intraclass correlation coefficient (3, 1) is 0.729, and the 95% confidence interval (CI) ranges from 0.331 to 0.999. Both analyses provide evidence for high test–retest reliability of our scale. Hence, we strongly believe that the phrasing of the items does not affect our results. Inverting the responses of the positively phrased items offers a reliable measure of NIH.

TABLE 2 Study 2: Final version of the explicit NIH scale and factor loadings

Item		Factor loading
Affective subscale		
A1	I like to work with nonrelated or less related subject areas.	0.785
A2	I have sympathies for other knowledge domains.	0.786
A3	I look forward to talks and speeches from other knowledge domains.	0.767
Cognitive subscale		
C1	Collaborating with other knowledge domains generates more overhead than benefit.	0.611
C2	I think that different knowledge backgrounds may be helpful for the progress of a project.	0.621
C3	I doubt that I could achieve significant results applying methods taken from other knowledge domains.	0.554
Behavioral subscale		
B1	I network across different knowledge domains.	0.785
B2	I look for opportunities to exchange with persons having a different knowledge background.	0.704
B3	In addition to the challenges of my own discipline, I seek new ones at the interfaces to other disciplines.	0.680

Note. Measured with a slider on a metric scale ranging from 0 (*strongly disagree*) to 100 (*strongly agree*). All factor loadings in this table are significant below $p = .001$.

NIH, not invented here.

TABLE 3 Study 2: Coefficients from confirmatory factor analysis

	Affective component	Cognitive component	Behavioral component	Fornell–Larcker ratio	Composite reliability	Average variance extracted
Affective component	0.78			0.90	0.82	0.61
Cognitive component	0.69	0.60		1.15	0.62	0.36
Behavioral component	0.70	0.63	0.72	0.96	0.77	0.53

Note. The first three columns contain the correlation coefficients of the three subscales with the square root of the average variance extracted in the diagonal.

TABLE 4 Study 2: Fit statistics for proposed and alternative models

	Three component model	Alternative 1: Fixed correlation of cognitive and behavioral component	Alternative 2: Fixed correlation of cognitive and affective component	Alternative 3: Fixed correlation of affective and behavioral component	Alternative 4: One component model
χ^2	39.2	68.6	63.7	92.1	113.2
df	24	25	25	25	27
$\Delta\chi^2$ (Δdf)		29.4(1)***	24.5(1)***	52.9(1)***	74.0(3)***
χ^2/df	1.63	2.75	2.55	3.68	4.19
(S)RMR	0.04	0.06	0.06	0.06	0.07
RMSEA	0.05	0.09	0.09	0.11	0.12
CFI	0.98	0.93	0.94	0.89	0.86

Note. CFI, comparative fit index; RMSEA, root mean square error of approximation; (S)RMR, standardized root mean square residual; TLI, Tucker–Lewis index (=non-normed fit index).

$^{***}p < .0001$.

TABLE 5 Study 2: Means, standard deviations, and correlations of the different explicit measures

	Mean	Standard deviation	NIH score	Affective subscale	Cognitive subscale	Behavioral subscale
NIH score	33.96	12.47	1			
Affective subscale	29.32	14.31	0.89***	1		
Cognitive subscale	31.61	12.51	0.81***	0.61***	1	
Behavioral subscale	40.95	16.40	0.89***	0.70***	0.54***	1

Note. NIH, not invented here.

$^{***}p < .0001$.

had to find novel appropriate stimuli for the latter category. Following the process described before, 15 individuals specified terms that represent the management discipline. The consolidated list incorporated more than 30 terms. Next, 21 individuals rated all of these words using a 5-point Likert scale. We used the 10 words with the highest average rating for the study (e.g., controlling, revenue, and marketing). To represent mechanical engineering and the valence categories, we randomly drew 10 words from each of the lists discussed above.

5.1.4 | IAT procedure

Following the general design of a seven-block IAT (Greenwald et al., 2003), all blocks comprised 20 trials, except Blocks 5 and 7, which contained 40 trials. In the mixed blocks (20 trials) including both knowledge domains and attribute categories, stimuli were drawn randomly for each category, ensuring that every category was presented with an equal number of stimuli. Again, the category and attribute label combinations and the placing on the screen (left or right) were balanced across participants. Due to our experience from the first study, we focused on the D measure as the IAT effect measure. We found a good split-half reliability (Spearman–Brown coefficient) of 0.84 for the D measure.

5.2 | Results

5.2.1 | Explicit NIH scale

In line with Study 2, we analyzed the three subscales and the overall NIH score. Table 6 lists means and standard deviations for all measures. Taking the range of the scales into account, participants reported moderate NIH scores. The three different subscales are again strongly correlated. We calculated a number of *t* tests to inspect the differences between the two groups that had a background in either mechanical engineering or management. We found significant differences for all scales except the behavioral subscale, all showing higher NIH scores for the management students than for mechanical engineering students (Table 7).

5.2.2 | Implicit-association test

In line with Study 1, the participants showed a negative effect ($\mu = 0.434, \sigma = 0.527$) towards the respective other knowledge domain, $t(123) = 9.176$, $p < .001$, $r = .637$. Interestingly, we found a large difference in the IAT effect between the management students ($\mu = 0.724, \sigma = 0.382$) and mechanical engineering students ($\mu = 0.154, \sigma = 0.500$), $t(122) = -7.15$, $p < .001$, $r = .543$. The order of the IAT test sequences and categories had no significant influence on the D measure, $F(3, 120) = 0.217$, ns.

5.2.3 | IAT scale correlation

To investigate the convergence of the IAT and the explicit survey measure for NIH, we calculated the correlations of the D measure, the NIH survey score, and the scores of the three subscales. As displayed in Table 6, all correlations are in the expected positive direction and show a significant correspondence within the different attitude measures. Notably, the correlation with the behavioral subscale is smaller.

5.3 | Discussion

Our third study examined the convergence of explicit and implicit association-based attitude measures. Both instruments indicated the presence of NIH tendencies in our sample.[6] Interestingly, the scale scores exhibited modest attitude values, whereas the D measure calculated on the IAT data revealed rather strong effects. This may be interpreted in three different ways. First, the implicit and explicit cognitions underlying the depicted attitudes may differ. This would mean that individuals might hold different attitudes towards external input that may drive behavior in an implicit or explicit manner. Second, the difference could reflect a lack of introspection into the underlying NIH attitude; the individuals might have a negative attitude towards external input, which is reflected by strong IAT effects, but they lack introspection in that attitude and thus report a more positive self-perception on the explicit scale. Third, the smaller effects measured by the explicit scale might be due to social desirability. Rather than lacking introspection, individuals might have reported euphemized attitudes on the scales, which was revealed by the IAT.

The correlation results favor the first interpretation. Although the correlation coefficient between the score of the (explicit) NIH scale and the (implicit) D measure drawn from the IAT ($r = .340$) is rather modest with regard to common standards, it provides evidence for convergent validity of both measures. The correlation exceeds the mean implicit–explicit correlations of recent meta-analytic studies on the IAT reporting a mean correlation coefficient of 0.24 (Greenwald et al., 2009; Hofmann et al., 2005). As the coefficient at hand exceeds meta-analytic results and the IAT is an implicit measure without allowing insight into the measured construct, it is unlikely that the study participants faked responses. If it is true that the different effect sizes indicate different explicit and implicit NIH cognitions, then they seem to be more alike than average because they exceed the meta-analytic correlation. Scholars interested in measuring NIH in their own study might argue that due to the relatively strong correlation in comparison with that of other attitude studies and the additional effort of administrating an implicit measure, it is appropriate to apply a standard scale measure. However, this additional effort needs to be taken in case of a specific (theoretical) interest in the implicit component of the NIH attitude.

6 | STUDY 4—DISCRIMINANT AND CRITERION VALIDITY OF THE NIH SCALE

Study 3 found that a standard scale might be appropriate for most measurements of NIH in an organizational context. This fourth study aims at enhancing the evidence for validity of this direct NIH measure.[7] We designed the study to further investigate construct validity

[6]Interestingly, the participant groups differed with respect to both implicitly and explicitly measured attitude values. We are very cautious to state that individuals with a background in mechanical engineering have weaker attitudes towards other knowledge domains. In fact, participants were aware that they signed up to an experiment taking place at the School of Business and Economics. We assume that individuals with a background in mechanical engineering signing up were more open towards the management knowledge domain, and thus on average exhibited lower D measures.

[7]We thank one anonymous reviewer for inspiring us to conduct this study.

TABLE 6 Study 3: Means, standard deviations, and correlations of the different attitude measures

	Mean	Standard deviation	NIH score	Affective subscale	Cognitive subscale	Behavioral subscale	IAT score
NIH score	32.65	13.35	1				
Affective subscale	31.21	16.11	0.51***	1			
Cognitive subscale	27.79	13.84	0.37***	0.58***	1		
Behavioral subscale	38.96	18.99	0.72***	0.86***	0.84***	1	
IAT score	0.43	0.53	0.34***	0.34***	0.32***	0.18*	1

Note. IAT, implicit-association test; NIH, not invented here.

***$p < .0001$.

TABLE 7 Study 3: Differences between participants having a background in management and ME

	Mean (standard deviation) of management students	Mean (standard deviation) of mechanical engineering (ME) students	*t(df), r*
NIH score	35.13 (14.43)	41.74 (12.15)	−2.81 (128)**, 0.24
Affective subscale	35.09 (17.50)	42.65 (16.38)	−2.54 (128)*, 0.22
Cognitive subscale	27.68 (13.06)	35.41 (13.83)	−3.28 (128)**, 0.28
Behavioral subscale	42.64 (21.83)	47.17 (16.08)	−1.34 (128)

Note. NIH, not invented here.

**$p < .01$; *$p < .05$.

of our direct NIH scale by looking at its discriminant validity. In addition, we investigate its criterion validity.

6.1 | Method

6.1.1 | General procedure and participants

We collected contact details of 3,643 German scientists with a background in the Sciences, Technology, Engineering, and Mathematics disciplines working for one of Germany's largest research institution. Of these, 826 individuals agreed to participate in our survey. We asked the participants to think of one interdisciplinary research project conducted throughout the last year. We instructed them to answer all questions of the survey with regard to this specific project. We received 370 completed responses (response rate 44.8%).

6.1.2 | Variables to establish discriminant validity

In order to establish discriminant validity of our NIH scale reported in Table 2, we included two constructs investigated in prior research that are related to but distinct from NIH on a conceptual level. *Individual learning orientation* has been investigated as a prerequisite to employee's creative output and is defined as "a concern for, and dedication to, developing one's competence" (Gong, Huang, & Farh, 2009, p. 765). We relied on an established scale to measure the construct (Elliot & Church, 1997; Gong et al., 2009). In a similar vein, a large body of research exists on intergroup biases (Hewstone, Rubin, & Willis, 2002). This research investigates attitudes and stereotypes of individuals towards members of other groups and is conceptually distinct to NIH as the latter is an attitude towards knowledge content rather than the individuals owning or developing the content. We relied on Huff and Kelley's (2003) measure of *group bias*.

6.1.3 | Variables to establish criterion validity

Following Antons and Piller (2015), the attitude underlying NIH results in disturbed knowledge absorption that, in turn, affects project outcomes and success. To measure *knowledge absorption*, we used four items ("I readily adopted the knowledge (data, suggestions, technical know-how, guidance,...) provided by the external project partner," "I successfully integrated the knowledge provided by the partner into my task," "I tried to gain as much knowledge as possible from the partner's expertise," and "I made sure to thoroughly understand everything the partner explained to me"; Cronbach's α = 0.73). We measured *project success* on the basis of the items by Dvir and Lechler (2004).

We controlled for a number of project-related attributes (degree of innovativeness, the technical proximity of the project partners, the project's complexity, and project foci including product, service, process, and technology development) and a set of individual attributes (gender, age, and tenure).

6.2 | Results

6.2.1 | Discriminant validity

Using AMOS to calculate a confirmatory factor analysis, we put three different latent constructs into the structural model to represent the three different attitude components of NIH (affective, cognitive, and behavioral components). In addition, we introduced latent constructs representing group bias and individual learning orientation. Illustrated in Table 8, the Fornell–Larcker test revealed that the cognitive component might have insufficient discriminant validity (Fornell & Larcker, 1981). We tested four alternative models to compare the fit indices as an alternative test (Byrne, 2010). With the high correlations of the cognitive component with the affective and the behavioral component of the attitude as well as group bias, we introduced models where the

TABLE 8 Study 4: Coefficients from confirmatory factor analysis

	Affective component	Cognitive component	Behavioral component	Individual learning orientation	Group bias	Fornell–Larcker ratio
Affective component	1					0.906
Cognitive component	0.628	1				1.733
Behavioral component	0.565	0.535	1			0.565
Individual learning orientation	0.262	0.265	0.364	1		0.326
Group bias	−0.317	−0.513	−0.185	0.007	1	0.377

Note. The first five columns contain the correlation coefficients of the scales.

correlation between those latent constructs is fixed (Alternatives 1–3). In addition, we introduced one model that contained only one latent factor for the NIH attitude in combination with the two other latent constructs. As depicted in Table 9, these tests provide evidence for discriminant validity of our NIH scales: The model containing five different latent constructs possesses a good model fit with regard to the fit indices (Byrne, 2010; Hu & Bentler, 1999; Schermelleh-Engel et al., 2003; Sharma et al., 2005). The initial model has better indices than all alternatives, and the alternatives lead to significant decreases in model fit.

6.2.2 | Criterion validity

As Antons and Piller (2015) discuss, NIH attitudes should bias individuals' behavior during research projects so that they reject external knowledge and this, in turn, should affect project performance. Only then will NIH attitudes result in negative outcomes for the organization by distorting knowledge absorption. We find support for this conceptual model. Table 10 reports our regression results. Models 1 and 2 serve as baseline models only including the sets of control variables. Model 3 introduces NIH into the regression of knowledge absorption, indicating a negative association of NIH and the degree of absorption of the interdisciplinary partner's knowledge (b = −0.352, p < .01). Model 4 tests the association of NIH and project success, yielding a negative coefficient (b = −0.235, p < .01).[8] Finally, Model 5 finds a partial mediation. A bootstrap test with 1,000 replications indicates the significance of the indirect effect of NIH on project success mediated by its negative association with knowledge absorption (−.050, bias-corrected 95% CI [−0.103, −0.009]).

6.3 | Discussion

Study 4 deepens our knowledge on the psychometrics of our developed NIH scale. It provides evidence for discriminant validity on the basis of investigations of two related but conceptually distinct constructs. Together with the results of Study 3, this provides evidence for construct validity of the measure. Moreover, we investigated criterion validity by looking at concurrent validity of behavioral (knowledge absorption) and outcome measures (project success) of interdisciplinary research projects. Although our approach clearly is not suited to identify NIH as a causal driver of these outcomes, we provide the first empirical evidence for the assumption that an NIH attitude exists and is associated to biased knowledge absorption and distorted project success.

7 | GENERAL DISCUSSION

The role of NIH as an obstacle for knowledge transfer and absorption, as well as for individual and organizational learning, has been widely acknowledged (Antons & Piller, 2015; Franke et al., 2009; Gesing et al., 2015; Hauser et al., 2006; Laursen & Salter, 2006). This calls for a better understanding of NIH's underlying mechanisms, its impact, and possible countermeasures. To conduct this kind of research, scholars need appropriate measures of the phenomenon. So far, NIH has been operationalized in various and sometimes rather obscure ways, which have been criticized due to their lack of reliability and validity (Agrawal et al., 2010; Antons & Piller, 2015). Introducing and applying theoretical findings from psychological attitude research, we developed and applied an implicit measure as well as an explicit measure of NIH. We provided evidence for convergent validity of both measures as well as the discriminant and criterion validity of the explicit measure. These measurement instruments for NIH will hopefully contribute to a better understanding of NIH in future studies, facilitating better knowledge transfer and exchange across different disciplines, organizational units, and geographic borders.

7.1 | Theoretical implications

We believe that our research advances the literature on NIH in at least five meaningful ways. First, we introduce the tripartite componential model of attitudes (Rosenberg & Hovland, 1960) as a theoretical framework to better ground measurement development for the NIH attitude. Moreover, we distinguish between implicit and explicit social cognitions (Fazio & Olson, 2003; Greenwald & Banaji, 1995), which provides the theoretical foundation for the development of both implicit and explicit measures. Prior NIH research was grounded in various theoretical frameworks, such as social identity (Agrawal et al., 2010; Allen et al., 1988; Lichtenthaler & Ernst, 2006), cultural perspectives (de Burcharth et al., 2014; Kanter, 1983), group routines, and the attempt to reduce uncertainty in decision making (Kathoefer & Leker, 2012; Katz & Allen, 1982). The development of measures for NIH in these studies has often proceeded somewhat

[8]Following a comment from an anonymous reviewer, we replicated the regression analyses using the three different attitude components as predicators instead of the NIH construct. This yielded negative associations between all components and knowledge absorption and project success. Variance inflation tests revealed that the models did not suffer from multicollinearity, which supports that the developed measures are sufficiently discriminant. Finally, the regression coefficients did not differ significantly with respect to their magnitude.

TABLE 9 Study 4: Fit statistics for proposed and alternative models

	Model containing five constructs	Alternative 1: Fixed correlation of cognitive and affective component	Alternative 2: Fixed correlation of cognitive and behavioral component	Alternative 3: Fixed correlation of cognitive component and group bias	Alternative 4: One latent construct for NIH
χ^2	239.8	272.4	279.1	438.4	408.0
df	125	126	126	126	132
$\Delta\chi^2$ (Δdf)		32.6(1)***	39.3(1)***	198.6(1)***	168.2(7)***
χ^2/df	1.918	2.162	2.215	3.479	3.091
(S)RMR	0.06	0.08	0.08	0.13	0.07
RMSEA	0.05	0.06	0.06	0.08	0.08
CFI	0.95	0.93	0.93	0.85	0.87

Note. CFI, comparative fit index; NIH, not invented here; RMSEA, root mean square error of approximation; (S)RMR, standardized root mean square residual; TLI, Tucker–Lewis index.

TABLE 10 Study 4: Results from OLS regression

		Model 1 Knowledge adoption	Model 2 Project success	Model 3 Knowledge adoption	Model 4 Project success	Model 5 Project success
1.	Constant	3.744*** (0.297)	3.798*** (0.296)	4.957*** (0.346)	4.608*** (0.354)	3.900*** (0.440)
Control set: Project attributes						
2.	Innovativeness	0.018 (0.053)	0.104** (0.053)	−0.000 (0.051)	0.092* (0.052)	0.092* (0.051)
3.	Technical proximity	0.208*** (0.031)	0.171*** (0.031)	0.211*** (0.029)	0.172*** (0.030)	0.142*** (0.032)
4.	Complexity	0.147*** (0.038)	0.027 (0.038)	0.131*** (0.037)	0.016 (0.038)	−0.003 (0.038)
5.	Product focus	0.017 (0.093)	−0.096 (0.092)	−0.003 (0.088)	−0.109 (0.090)	−0.108 (0.090)
6.	Process focus	−0.152 (0.095)	0.039 (0.095)	−0.150* (0.091)	0.040 (0.093)	0.062 (0.093)
7.	Service focus	−0.291** (0.115)	0.093 (0.115)	−0.319*** (0.110)	0.075 (0.112)	0.121 (0.113)
8.	Technology focus	−0.146 (0.105)	−0.042 (0.104)	−0.182* (0.100)	−0.066 (0.102)	−0.040 (0.102)
Control set: Individual attributes						
9.	Gender	0.133 (0.112)	0.090 (0.111)	0.114 (0.106)	0.077 (0.109)	0.061 (0.108)
10.	Age	0.002 (0.004)	0.004 (0.004)	0.000 (0.003)	0.002 (0.003)	0.002 (0.003)
11.	Tenure	0.001 (0.003)	0.004 (0.003)	0.001 (0.003)	0.004 (0.003)	0.004 (0.003)
Main effects						
12.	NIH			−0.352*** (0.058)	−0.235*** (0.059)	−0.185*** (0.061)
Mediation						
13.	Knowledge adoption					0.143*** (0.054)
Observations	370	370	370	370	370	
F	8.11***	5.59***	11.52***	6.74***	6.88***	
R^2	0.184	0.135	0.261	0.1717	0.1879	
Adjusted R^2	0.162	0.111	0.239	0.1462	0.1606	

Note. Unstandardized estimates are reported. Standard errors are reported in parentheses.

NIH, not invented here; OLS, ordinary least squares.

***$p < .01$;

**$p < .05$;

*$p < .10$.

superficially, without thorough theoretical grounding. Given the broad empirical evidence provided in psychological literature, our understanding of explicit and implicit social cognition, as well as the tripartite componential model, holds potential to inform research in innovation management on NIH by investigating antecedents of the phenomenon related to each of its structural dimensions.

Beyond that, it offers potential to advance a broad range of related topics in organizational studies focusing on attitudes and stereotypes.

Second, our paper introduces an implicit measure of NIH through a newly adapted IAT (Greenwald et al., 1998), which is part of the association-based category of implicit measures (Uhlmann et al., 2012) and the most prevalent implicit measure (Nosek et al., 2011). In Study 1, we provided an introduction to the development of stimuli and materials for IAT applications in management research. We explained the procedure of the IAT, its criterion, calculation, and analysis. This showcased the adaption of the IAT to a particular research question in innovation management, which also bears plenty of opportunities for further research beyond NIH in organizational studies (e.g., Derous et al., 2012; Harms & Luthans, 2012; Leavitt et al., 2011). In general, association-based measures such as the IAT are well suited to answer questions with regard to assessing evaluations and predicting behavior or decisions (Greenwald et al., 2009; Nosek, Greenwald, & Banaji, 2007; Uhlmann et al., 2012). The ability to assess individual differences, coupled with its high predictive validity, provides opportunities to deepen our knowledge of implicit processes influencing individual actions in organizations. One reason behind the high predictive validity is the limited opportunity for participants to purposively influence the outcome. Put differently, the attitude measured is not biased by participants deliberately responding in, for example, a more socially acceptable way. There are also more pragmatic benefits to implicit measures such as the IAT, such as the fact that fewer participants are needed as compared to conducting a study based on explicit measures. Especially in experimental studies, this entails a logistic and financial benefit of implicit measures over explicit measures.

Third, by applying standard scale development procedures (DeVellis, 2011; Hinkin, 1995, 1998), we also make available a validated explicit scale to measure NIH, addressing the common method of empirical investigation in management research. Compared to recent attempts to equip researchers with survey measures of NIH (de Burcharth et al., 2014; Kathoefer & Leker, 2012), our scale development is deeply rooted in attitude theory (Ajzen, 2001; Bohner & Wänke, 2002; Eagly & Chaiken, 1993; Rosenberg & Hovland, 1960). In Study 2, we established reliability as well as discriminatory validity of the three subscales. By doing so, our scale development provides additional evidence for the tripartite componential structure of attitude and supports the common view that evolved from the broad discussion on attitude structure (Bohner & Wänke, 2002; Breckler, 1984; Eagly & Chaiken, 1993).

Fourth, we provide evidence for construct and criterion validity of the NIH measures. Study 3 yielded evidence for the convergent validity of our explicit and implicit NIH measures. Although methodological research has long called for the establishment of construct validity by proving both discriminatory and convergent validity of measures by means of multitrait–multimethod studies and comparisons of different measures for constructs (Campbell & Fiske, 1959; Hinkin, 1995), scale development in management research—even of well-known and highly cited scales—often falls short in this regard (e.g., Flatten, Engelen, Zahra, & Brettel, 2011; Liden & Maslyn, 1998). Study 4 added by evidence on the discriminant validity of the NIH scale and measures of other related construct. Finally, we are among the first showing that the NIH attitude is really associated with the formerly postulated negative outcomes in innovative endeavors. NIH is associated with disturbed adoption of external knowledge content, which partially mediates NIH's link to derogated project success.

Fifth, innovation management scholars and other disciplines interested in NIH are now equipped with measures to assess NIH, making it detectable and manageable. NIH is detrimental to approaches opening up innovation processes to external contributors such as customers, suppliers, and universities—which is one of the most discussed topics in innovation research (Antons, Kleer, & Salge, 2016). Innovation research is now able to investigate formerly postulated antecedents of NIH (Agrawal et al., 2010; Antons & Piller, 2015; Herzog & Leker, 2010). Although we show that NIH is truly related to negative outcomes in innovation settings, our findings might be complemented by future studies looking at covariates of NIH fostering or overcoming its negative impact. Being able to measure NIH can also help to solve empirical puzzles in innovation literature, such as the hypothesis on complementary versus substitutional effects of internal and external R&D capacities (Gesing et al., 2015; Laursen & Salter, 2006). Being equipped with NIH measures, researchers might supplement existing measures of absorptive capacity with NIH measures to differentiate between their opposing effects (Antons & Piller, 2015).

7.2 | Managerial implications

Our study also has implications for management practice. First of all, managers should be aware of the habitual presence of NIH when dealing with external knowledge. The good news is that NIH can be measured and managers can, at least, try to address it given its prevailing extent. The instruments presented in this paper equip managerial practice with helpful measurements that enable an organization to evaluate the attitudes of its members towards external input. Applying these tools allows managers not just to evaluate the individual preposition of a staff member (a practice that in many organizations, however, may not be compliant with labor laws or employment agreement), but especially to assess on a team or department level whether a group is more or less afflicted with the NIH syndrome. In projects where external input is crucial, team selection would benefit from having knowledge about staff openness. Besides testing the existing workforce, the measures may also be useful when screening potential new hires (again, this application might be regulated in some countries). Another option is to offer the NIH measurement tools for (anonymous) self-assessment so that individuals can self-reflect on their attitudes towards external input and their consequent actions. Given the results of our empirical analyses that (a) both implicit and explicit approaches produce reliable and converging results and (b) the NIH scale possesses discriminant and criterion validity, we recommend the application of explicit scales, as they are easier to execute and closer to the experience of an employee.

When management is able to measure NIH, it can test the effectiveness of remedies. At this point, NIH starts becoming controllable. Research on attitudes has shown that, for example, perspective taking or persuasive communication can change attitudes (Bohner & Dickel, 2011; Todd & Burgmer, 2013). Such techniques help to deepen individual introspection and allow reflection on one's own attitudes towards external input stemming from other backgrounds. Because reflection is seen as a major driver of attitude change, we recommend using these techniques and, additionally, educating team members towards a better understanding of NIH. Enabling staff to reflect might be a very efficient means of changing and altering attitudes in a way that is more favorable from an organizational perspective. Explicit measurement can then be deployed to ensure a positive change after treatment.

7.3 | Limitations and avenues for future research

Our research is not without limitations. From a conceptual point of view, future research on attitudes should dive deeper into the question how the cognitive, affective, and behavioral components of an attitude are linked to the explicit and implicit distinction. Psychological reviews on attitudes highlight that attitude research is still in its infancy in understanding whether explicit and implicit measures tap into the same attitude, different aspects or dimensions of attitudes, or different processes activating attitudes (Bohner & Dickel, 2011; Crano & Prislin, 2006; Fazio & Olson, 2003).

From a methodological point of view, we relied to a large extent on student samples to develop our measures. As meta-analyses have shown (Kraus, 1995; Peterson, 2001), effects from student samples often differ from real life samples. However, as Kardes (1996) points out, student samples are reliable for studies that focus on general psychological processes and on identification of psychological mechanisms. In our case, the development of attitude measures fulfills this condition. Moreover, we were able to show in Study 1 that the samples of students and professionals did not differ with regard to the IAT effects. Finally, we did not rely on a student sample to test criterion validity of our NIH scale (the kind of study where meta-analyses have found differences between student and real life samples) but draw on data from a sample of 370 academics being employed as researchers by a large R&D institution.

Although the application of implicit measures holds great promise, scholars have to acknowledge that it usually takes participants more time and effort to complete such a test than standard survey measures. This often holds the risk of high dropout rates. Recent research has developed short (Sriram & Greenwald, 2009) and web-based versions (Greenwald et al., 2003; Nosek, Banaji, & Greenwald, 2002) of the IAT. Future methodological research in management should focus on the development of implicit measures, which are easily applicable in organizational settings. Here, it has to be noted that implicit association-based measures such as the IAT are by no means the best solution for every research question regarding implicit social cognition. Relying on very fast reactions, they are well suited to assess general tendencies, such as attitudes to a broad target category such as sex, race, or, as shown above, a knowledge domain. When interested in highly nuanced targets such as a single category in a very special and distinct situational context, scholars should favor other implicit methods over association-based ones (Uhlmann et al., 2012).

Notably, a few studies from different organizational fields have started to adopt implicit measures. Studies in consumer research and marketing have investigated implicit brand attitudes, prejudices against service employees, and ethical perceptions (Hekman et al., 2010; Luchs, Naylor, Irwin, & Raghunathan, 2010; Maison, Greenwald, & Bruin, 2004). Scholars in human resources have investigated implicit biases in hiring processes (Agerström & Rooth, 2011; Derous, Nguyen, & Ryan, 2009; Derous et al., 2012; Rooth, 2010; Rudman & Glick, 2001; Yogeeswaran & Dasgupta, 2010; Ziegert & Hanges, 2005). Similarly, some studies have applied implicit measures to understand antecedents of employee behavior and individual decision-making processes at work, for example, individual preferences for careers and jobs (Rudman & Heppen, 2003), prejudices influencing intentions to leave a job (von Hippel, Brener, & von Hippel, 2008), morality and decisions in business (Marquardt & Hoeger, 2009; Reynolds, Leavitt, & DeCelles, 2010), and job attitudes and behavior at work (Leavitt et al., 2011). These examples may stimulate ideas for further studies aiming to understand implicit processes regarding attitudes, stereotypes, self-concepts, and other individual social cognitions and their respective organizational outcomes.

Regarding NIH, Antons and Piller (2015) have highlighted that developing measures for the NIH attitude is the necessary precondition to investigating antecedents, consequences, contexts, and remedies for NIH. They have developed an extensive agenda for future research in these fields. Our research enables scholars to transform this agenda into actual research and knowledge. Moreover, organizational behavior research might consider answering the question whether NIH is a phenomenon only occurring in an innovation context. Inventing and innovating are by definition knowledge-intensive tasks. It might be likely that other professions that are characterized by their knowledge intensity, such as physicians, legal consultation, or financial auditing, are also affected by negative attitudes towards external knowledge.

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APPENDIX A

TABLE A1 IAT scoring algorithms

The conventional scoring algorithm recommended by Greenwald et al. (1998)	
1	Exclude subjects with excessively slow responding and/or high error rates
2	Drop the first two trials of each block
3	Recode latencies faster than 300 to 300 ms
4	Recode latencies slower than 3,000 ms to the 3,000-ms boundary
5	Log transform all latency values
6	Compute the mean for the two test blocks
7	Compute the difference of the two means
For the five-block IAT, Step 6 is conducted for Blocks 3 and 5. For the seven-block IAT, this step is proceeded for Blocks 4 and 7 accordingly.	
The improved scoring algorithm recommended by Greenwald et al. (2003)	
1	Delete trials greater than 10,000 ms
2	Delete subjects for whom more than 10% of all trials have latency less than 300 ms
3	Compute the pooled standard deviation for all trials in Blocks 3 and 6 and accordingly for Blocks 4 and 7
4	Compute the mean of the correct latencies for each block
5	Replace each error latency with block mean + 600 ms
6	Compute the mean of the resulting values for each block
7	Compute two differences: Mean of Block 6 minus mean of Block 3 and accordingly for Blocks 7 and 4
8	Divide each difference score by its associated pooled standard deviation
9	Compute the mean of the two resulting values
To adapt the algorithm to the five-block IAT, we skipped steps 3 and 7 and divided the difference of the block means by the pooled standard deviation for both test bocks.	

APPENDIX B

TABLE B1 Study 2: Item analysis

Iteration and participants		1 (316)		2 (124)	
Attitude component	**Criterion**	**Initial item set**	**Revised item set**	**Initial item set**	**Revised item set**
Affective component	Item difficulty	0.75	0.72	0.71	0.73
	Item-to-total correlation	0.49	0.57	0.53	0.63
	Item homogeneity	0.31	0.42	0.37	0.49
	Cronbach's α	0.80	0.81	0.76	0.85
Cognitive component	Item difficulty	0.66	0.70	0.65	0.68
	Item-to-total correlation	0.33	0.39	0.34	0.43
	Item homogeneity	0.16	0.22	0.18	0.29
	Cronbach's α	0.70	0.69	0.65	0.69
Behavioral component	Item difficulty	0.66	0.65	0.61	0.61
	Item-to-total correlation	0.29	0.27	0.45	0.52
	Item homogeneity	0.18	0.22	0.28	0.38
	Cronbach's α	0.51	0.44	0.73	0.75